

XIST RNA: a window into the broader role of RNA in nuclear chromosome architecture.

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XIST RNA triggers the transformation of an active X chromosome into a condensed, inactive Barr body and therefore provides a unique window into transitions of higher-order chromosome architecture. Despite recent progress, how XIST RNA localizes and interacts with the X chromosome remains poorly understood. Genetic engineering of XIST into a trisomic autosome demonstrates remarkable capacity of XIST RNA to localize and comprehensively silence that autosome. Thus, XIST does not require X chromosome-specific sequences but operates on mechanisms available genome-wide. Prior results suggested XIST localization is controlled by attachment to the insoluble nuclear scaffold. Our recent work affirms that scaffold attachment factor A (SAF-A) is involved in anchoring XIST, but argues against the view that SAF-A provides a unimolecular bridge between RNA and the chromosome. Rather, we suggest that a complex meshwork of architectural proteins interact with XIST RNA. Parallel work studying the territory of actively transcribed chromosomes suggests that repeat-rich RNA 'coats' euchromatin and may impact chromosome architecture in a manner opposite of XIST. A model is discussed whereby RNA may not just recruit histone modifications, but more directly impact higher-order chromatin condensation via interaction with architectural proteins of the nucleus. This article is part of the themed issue 'X-chromosome inactivation: a tribute to Mary Lyon'.

Establishment and maintenance of random monoallelic expression.

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This Review elucidates the regulatory principles of random monoallelic expression by focusing on two well-studied examples: the X-chromosome inactivation regulator Xist and the olfactory receptor gene family. Although the choice of a single X chromosome or olfactory receptor occurs in different developmental contexts, common gene regulatory principles guide monoallelic expression in both systems. In both cases, an event breaks the symmetry between genetically and epigenetically identical copies of the gene, leading to the expression of one single random allele, stabilized through negative feedback control. Although many regulatory steps that govern the establishment and maintenance of monoallelic expression have been identified, key pieces of the puzzle are still missing. We provide an overview of the current knowledge and models for the monoallelic expression of Xist and olfactory receptors. We discuss their similarities and differences, and highlight open questions and approaches that could guide the study of other monoallelically expressed genes.

LncRNA XIST/miR-455-3p/HOXC4 axis promotes breast cancer development by activating TGF- β /SMAD signaling pathway.

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Breast cancer is the second primary cause of cancer death among women. Long non-coding RNA (lncRNA) X-inactive specific transcript (XIST) is a central regulator for X chromosome inactivation, and its abnormal expression is a primary feature of breast cancer. So far, the mechanism of XIST in breast cancer has not been fully elucidated. We attempted to illustrate the mechanism of XIST in breast cancer. The expressions of XIST, microRNA-455-3p (miR-455-3p) in breast cancer were measured using

quantitative real-time PCR. The expressions of homeobox C4 (HOXC4) were assessed with immunohistochemical and Western blot. Also, the functions of XIST in breast cancer were assessed by Cell Counting Kit-8 analysis, colony formation assay, flow cytometry, Western blot, Transwell, and cell scratch assays. Meanwhile, the mechanism of XIST in breast cancer was validated using database analysis and dual-luciferase reporter assay. Furthermore, the function of XIST in breast cancer in vivo was estimated by tumor xenograft model, immunohistochemical assay, and hematoxylin-eosin staining. XIST and HOXC4 expressions were increased, but miR-455-3p expressions were decreased in breast cancer tissues and cells. Knocking down XIST restrained breast cancer cell proliferation, invasion, migration, epithelial-mesenchymal transformation (EMT), and induced cell cycle arrest at G0/G1. Meanwhile, XIST interacted with miR-455-3p, while miR-455-3p interacted with HOXC4. XIST knockdown repressed breast cancer cell proliferation, invasion, and EMT, while miR-455-3p inhibitor or HOXC4 overexpression abolished those impacts. HOXC4 overexpression also blocked the impacts of miR-455-3p mimic on breast cancer cell malignant behavior. In vivo experimental data further indicated that XIST knockdown repressed breast cancer cell tumorigenic ability, and decreased HOXC4 and p-SMAD3 (TGF- β /SMAD-related protein) expressions. XIST/miR-455-3p/HOXC4 facilitated breast cancer development by activating the TGF- β /SMAD pathway.

Identification of the RSX interactome in a marsupial shows functional coherence with the Xist interactome during X inactivation.

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The marsupial specific RSX lncRNA is the functional analogue of the eutherian specific XIST, which coordinates X chromosome inactivation. We characterized the RSX interactome in a marsupial representative (the opossum *Monodelphis domestica*), identifying 135 proteins, of which 54 had orthologues in the XIST interactome. Both interactomes were enriched for biological pathways related to RNA processing, regulation of translation, and epigenetic transcriptional silencing. This represents a remarkable example showcasing the functional coherence of independently evolved lncRNAs in distantly related mammalian lineages.

Xist RNA and the mechanism of X chromosome inactivation.

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Dosage compensation in mammals is achieved by the transcriptional inactivation of one X chromosome in female cells. From the time X chromosome inactivation was initially described, it was clear that several mechanisms must be precisely integrated to achieve correct regulation of this complex process. X-inactivation appears to be triggered upon differentiation, suggesting its regulation by developmental cues. Whereas any number of X chromosomes greater than one is silenced, only one X chromosome remains active. Silencing on the inactive X chromosome coincides with the acquisition of a multitude of chromatin modifications, resulting in the formation of extraordinarily stable facultative heterochromatin that is faithfully propagated through subsequent cell divisions. The integration of all these processes requires a region of the X chromosome known as the X-inactivation center, which contains the Xist gene and its cis-regulatory elements. Xist encodes an RNA molecule that plays critical roles in the choice of which X chromosome remains active, and in the initial spread and establishment of silencing on the

inactive X chromosome. We are now on the threshold of discovering the factors that regulate and interact with Xist to control X-inactivation, and closer to an understanding of the molecular mechanisms that underlie this complex process.

Gene silencing in X-chromosome inactivation: advances in understanding facultative heterochromatin formation.

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In female mammals, one of the two X chromosomes is silenced for dosage compensation between the sexes. X-chromosome inactivation is initiated in early embryogenesis by the Xist RNA that localizes to the inactive X chromosome. During development, the inactive X chromosome is further modified, a specialized form of facultative heterochromatin is formed and gene repression becomes stable and independent of Xist in somatic cells. The recent identification of several factors involved in this process has provided insights into the mechanism of Xist localization and gene silencing. The emerging picture is complex and suggests that chromosome-wide silencing can be partitioned into several steps, the molecular components of which are starting to be defined.

Nuclear mRNA degradation pathway(s) are implicated in Xist regulation and X chromosome inactivation.

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A critical step in X-chromosome inactivation (XCI), which results in the dosage compensation of X-linked gene expression in mammals, is the coating of the presumptive inactive X chromosome by the large noncoding Xist RNA, which then leads to the recruitment of other factors essential for the heterochromatinisation of the inactive X and its transcriptional silencing. In an approach aimed at identifying genes implicated in the X-inactivation process by comparative transcriptional profiling of female and male mouse gastrula, we identified the Eif1 gene involved in translation initiation and RNA degradation. We show here that female embryonic stem cell lines, silenced by RNA interference for the Eif1 gene, are unable to form Xist RNA domains upon differentiation and fail to undergo X-inactivation. To probe further an effect involving RNA degradation pathways, the inhibition by RNA interference of Rent1, a factor essential for nonsense-mediated decay and Exosc10, a specific nuclear component of the exosome, was analysed and shown to similarly impair Xist upregulation and XCI. In Eif1-, Rent1-, and Exosc10-interfered clones, Xist spliced form(s) are strongly downregulated, while the levels of unspliced form(s) of Xist and the stability of Xist RNA remain comparable to that of the control cell lines. Our data suggests a role for mRNA nuclear degradation pathways in the critical regulation of spliced Xist mRNA levels and the onset of the X-inactivation process.

Spatial orchestration of the genome: topological reorganisation during X-chromosome inactivation.

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Genomes are organised through hierarchical structures, ranging from local kilobase-scale cis-regulatory contacts to large chromosome territories. Most notably, (sub)-compartments partition chromosomes according to transcriptional activity, while topologically associating domains (TADs) define cis-regulatory landscapes. The inactive X chromosome in mammals has provided unique insights into the regulation and function of the three-dimensional (3D) genome. Concurrent with silencing of the majority of genes and major alterations of its chromatin state, the X chromosome undergoes profound spatial rearrangements at multiple scales. These include the emergence of megadomains, alterations of the compartment structure and loss of the majority of TADs. Moreover, the Xist locus, which orchestrates X-chromosome inactivation, has provided key insights into regulation and function of regulatory domains. This review provides an overview of recent insights into the control of these structural rearrangements and contextualises them within a broader understanding of 3D genome organisation.

Screen for reactivation of MeCP2 on the inactive X chromosome identifies the BMP/TGF- β superfamily as a regulator of XIST expression.

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Rett syndrome (RS) is a debilitating neurological disorder affecting mostly girls with heterozygous mutations in the gene encoding the methyl-CpG-binding protein MeCP2 on the X chromosome. Because restoration of MeCP2 expression in a mouse model reverses neurologic deficits in adult animals, reactivation of the wild-type copy of MeCP2 on the inactive X chromosome (Xi) presents a therapeutic opportunity in RS. To identify genes involved in MeCP2 silencing, we screened a library of 60,000 shRNAs using a cell line with a MeCP2 reporter on the Xi and found 30 genes clustered in seven functional groups. More than half encoded proteins with known enzymatic activity, and six were members of the bone morphogenetic protein (BMP)/TGF- β pathway. shRNAs directed against each of these six genes down-regulated X-inactive specific transcript (XIST), a key player in X-chromosome inactivation that encodes an RNA that coats the silent X chromosome, and modulation of regulators of this pathway both in cell culture and in mice demonstrated robust regulation of XIST. Moreover, we show that Rnf12, an X-encoded ubiquitin ligase important for initiation of X-chromosome inactivation and XIST transcription in ES cells, also plays a role in maintenance of the inactive state through regulation of BMP/TGF- β signaling. Our results identify pharmacologically suitable targets for reactivation of MeCP2 on the Xi and a genetic circuitry that maintains XIST expression and X-chromosome inactivation in differentiated cells.

XIST in Brain Cancer.

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Long non-coding RNAs (lncRNAs) make up the majority of the human genome. They are a group of small RNA molecules that do not code for any proteins but play a primary role in regulating a variety of physiological and pathological processes. X-inactive specific transcript (XIST), one of the first lncRNAs to be discovered, is chiefly responsible for X chromosome inactivation: an evolutionary process of dosage compensation between the sex chromosomes of males and females. Recent studies show that XIST plays a pathophysiological role in the development and prognosis of brain tumors, a heterogeneous group of neoplasms that cause significant morbidity and mortality. In this review, we explore recent advancements in the role of XIST in migration, proliferation, angiogenesis, chemoresistance, and evasion

of apoptosis in different types of brain tumors, with particular emphasis on gliomas.
